

Supplementary Material: What Does a Computer-Use Agent Reliability Score Actually Measure?

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Abstract

Optional supplementary material for the AAMAS 2027 main-track submission. Per the call, this file does not count against the eight-page main-text limit, reviewers need not consult it, and it is not an extended or corrected version of the paper. Protocol tables only. No new experiment, no new estimand, no pooled n . The main paper is self-contained.

ACM Reference Format:

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1 Study 2 per-task common support

Numbers already used in the main text; this table is the per-task expansion of the exploratory four-task slice ($n_{\text{eff}}=1$). It is not a ranking and is not the confirmatory selection test.

2 Study 3 ordering under repairs

The main-text claims use R-AGG (+8/−18) and ALL (+10/−20) only. The rows below are the same locked counterfactuals on the four-task support; FROZEN is the baseline, not a finding. R-CHAN inversion is an existence demonstration ($n_{\text{eff}}=1$), not a recommended scoring rule.

R-AGG’s mean gap shrinks while per-task disagreement grows: nonzero ΔSTS tasks go $1 \rightarrow 2$, and the two moves cancel (counterfactual +0.250, retrieval-f009 −0.333).

3 Signed-movement robustness (R-AGG)

+8/−18 is a census of the sealed 134-row corpus, not a superpopulation estimate. The unit is the task. Thirteen tasks contribute any of the 26 abstentions R-AGG releases; the paper’s 61-task denominator is not padded here. Point counts match the lock: R-AGG sensitivity $20 \rightarrow 28$, MATCH destroyed = 0, 26 abstentions decompose as +8 correct / +18 wrong.

Leave-one-task-out on those 13 tasks: $C \in [5, 8]$, $W \in [10, 18]$, $C - W \in [-12, -3]$. Every fold keeps $C > 0$, $W > 0$, and $C < W$ (13/13). No single task is necessary for the licensed imbalance. Largest shares: 3/8 new-correct on retrieval-f009; 8/18 new-wrong on counterfactual-f013.

ALL is not jackknifed as +10/−20. FROZEN→ALL does not uniquely reproduce that pair: 31 released abstentions become +5 match /

Table 1: Per-task values on the four-task common support. Δ is Flash minus GPT-5.5. Task IDs as registered.

Task	S^0_G	S^0_F	STSG	STSF	ΔS^0	ΔSTS
cf-f010	83	100	0.000	0.000	+17	0.000
pref-f014	0	84	0.000	0.000	+84	0.000
retr-f002	75	100	0.500	0.500	+25	0.000
retr-f009	40	100	0.500	0.333	+60	−0.167

Table 2: Flash-minus-GPT ΔSTS on the four-task support.

Config	STS GPT	STS Flash	ΔSTS	Outcome
FROZEN	0.2500	0.2083	−0.0417	baseline
R-AGG	0.3333	0.3125	−0.0208	narrowed mean
R-SCOPE	0.2083	0.1875	−0.0208	narrowed mean
R-CMP	0.2500	0.2083	−0.0417	unchanged
R-CHAN	0.0417	0.2083	+0.1667	inverted
ALL	0.0417	0.2708	+0.2292	inverted

+26 M4; MATCH destroyed = 16 (8 →M4, 8 →ABSTAIN); sensitivity $\Delta = -10$, M4 $\Delta = +20$, abstention $\Delta = -10$. Those are mechanical transitions, not a new estimand.

Reproducing tree: commit 7b1f5073a06948d3b2fb3556f664668c7f011160; extractor 3242c30. Code and lockfiles are in the supplementary zip. No ranking, no CI, no replacement for the census.

4 Reconstruction (G2)

Locked rule R , G1 groundedness 30/30, G2 calibration 9/30. MATCH $20 \rightarrow 13$, RECALL_MISS $39 \rightarrow 46$, ABSENT 61 unmoved. R is not revised. The sealed corpus was not opened. G2 is not a result in the main paper.

5 Comparative cohort gate

Pre-specified threshold: $n \geq 20$ independent task clusters, same construct, frozen evaluator, decided before the first trajectory. Slate: 28 new family tasks (not the C7 support, not preference, not sealed). Probe: inject/guest feasibility only; no agent, no STS. Wave A 9/12, Wave B 7/16, $n = 16 < 20$. FAIL. Comparative branch closed. Not a scientific headline.

6 What this supplement is not

Not a place to pool 24+57+134. Not a ranking. Not a recommended repair. Not a G2 rescue. Not an extended version of the main paper.



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